

*MA.7.AR.4.4***Benchmark****MA.7.AR.4.4 Given any representation of a proportional relationship, translate the representation to a written description, table or equation.**

Example: The written description, there are 60 minutes in 1 hour, can be represented as the equation $m = 60h$.

Example: Gina works as a babysitter and earns \$9 per hour. She would like to earn \$100 to buy a new tennis racket. Gina wants to know how many hours she needs to work. She can use the equation $h = \frac{1}{9}e$, where e is the amount of money earned, h is the number of hours worked and $\frac{1}{9}$ is the constant of proportionality.

Benchmark Clarifications:

Clarification 1: Given representations are limited to a written description, graph, table or equation.

Clarification 2: Instruction includes equations of proportional relationships in the form of $y = px$, where p is the constant of proportionality.

Connecting Benchmarks/Horizontal Alignment

- MA.7.AR.3.3
- MA.7.GR.2.2
- MA.8.AR.3.2, MA.8.AR.3.3, MA.8.AR.3.4

Terms from the K-12 Glossary

- Constant of Proportionality
- Proportional Relationships
- Rate
- Unit Rates

Vertical Alignment**Previous Benchmarks**

- MA.6.AR.1.1
- MA.6.AR.3.2, MA.6.AR.3.5
- MA.7.NSO.2.2, MA.7.NSO.2.2, MA.7.NSO.2.3
- MA.7.AR.3.1, MA.7.AR.3.2

Next Benchmarks

- MA.912.AR.2.2

Purpose and Instructional Strategies

In previous courses, students translated written descriptions into algebraic expressions and translated algebraic expressions into written descriptions. In grade 7 accelerated, students translate any representation of a proportional relationship to a written description, table or equation. Students will also extend this work to include linear relationships, write an equation in slope-intercept form from a written description, a table, or a graph. In Algebra 1, students write a linear two-variable equation to represent a relationship given by a variety of mathematical and real-world contexts.

- Instruction includes different ways of representing proportional relationships, such as tables, equations and graphs. Multiplying or dividing one quantity in a ratio by a particular factor requires doing the same with the other quantity in the ratio to maintain the proportional relationship. Graphing equivalent ratios create a straight line passing through the origin. The equations generated with the ratios will be unique in that they will follow the form of $y = px$.

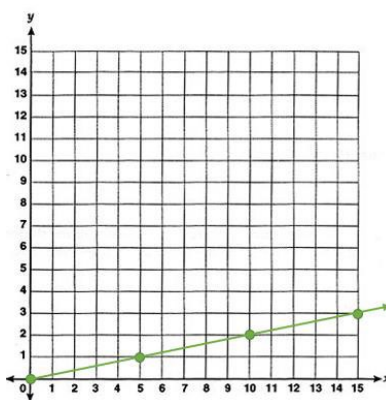
- Tables

x	5	40	65	
y	1	8	13	

- Equations

$$y = \frac{1}{5}x$$

- Graphs



- When providing a graph, be sure there are easily identifiable points for students to use in calculating the constant of proportionality.
- As students are building meaning, instruction makes connections between the different representations.
- Even though proportional relationships exist in Quadrant I, instruction includes opportunities for students to realize that the line does continue into Quadrant III but are not appropriate for the real-world situation.
- Students should be able to explain examples from the points on a graph or the numbers within the table by putting it back into the real-world context when appropriate.
- Instruction includes flexibility in understanding of the dependent and independent variables. Students can represent situations in terms of x or in terms of y .
 - For instance, within example 1 students can represent the situation as $m = 60h$ or $h = \frac{1}{60}m$.
- Students should construct verbal descriptions.
 - For example, a student might describe the situation as “the number of packs of gum times the cost for each pack is the total cost in dollars.” They can use the verbal model to construct the equation.

- Students can check the equation by substituting values and comparing their results to the table. The checking process helps students revise and recheck their model as necessary (*MTR.6.1*).
- Provide tables of values for various proportional relationships. Ask students to look at the tables and generalize how they can find the y -value in the tables given any x -value (*MTR.1.1*). Have students look for patterns and assist with developing the equation $y = px$ where p is the constant of proportionality (*MTR.5.1*).
- Ensure the formal development of the equation $y = px$ where p is the constant of proportionality. Instruction supports flexibility in the variable used for the constant of proportionality. Provide practice for students to develop this equation using different variables based on given scenarios, as in *Example 2*.

Common Misconceptions or Errors

- Students may neglect the scales on the axes when calculating and interpreting the constant of proportionality.
- Students may not be able to approximate the constant of proportionality from the graphs. To address this misconception, begin with graphs having easily identifiable points before moving toward problems that need approximations.
- Students may not see the connection between the constant of proportionality and the steepness of the graph. To address this misconception, provide a variety of graphs with various steepness and ask students to organize them based on increasing order of the constants of proportionality.

Strategies to Support Tiered Instruction

- Instruction includes utilizing the x - and y -axis when determining the constant of proportionality. Teacher provides instruction on locating the values for the variables y and x from the axis labels, rather than counting the minor gridlines to the chosen point on the graph.
- Instruction includes utilizing graphs containing easily identifiable points on minor gridlines before moving toward graphs containing points that lie between gridlines which requires estimation to determine an appropriate constant of proportionality.
- Instruction includes the co-creation of a graphic organizer containing examples of proportional relationships with increasing levels of steepness. For each example, include a real-world scenario, a table, a graph and the constant of proportionality.
- Instruction includes using letters for variables that relate to the given scenario, such as w for water.
- For students that are not able to approximate the constant of proportionality from the graphs, begin with graphs having easily identifiable points before moving toward problems that need approximations. For students that cannot see the connection between the constant of proportionality and the steepness of the graph, provide a variety of graphs with various steepness and ask students to organize them based on increasing order of the constants of proportionality.

Instructional Tasks

Instructional Task 1 (MTR.5.1)

Kell works at an after-school program at an elementary school. The table below shows how much money he earned every day last week.

	Monday	Wednesday	Friday
time worked	1.5 hours	2.5 hours	4 hours
money earned	\$12.60	\$21.00	\$33.60

Mariko has a job mowing lawns that pays \$7 per hour.

Part A. Who would make more money for working 10 hours? Explain or show work.

Part B. Draw a graph that represents y , the amount of money Kell would make for working x hours, assuming he made the same hourly rate he was making last week.

Part C. Using the same coordinate axes, draw a graph that represents y , the amount of money Mariko would make for working x hours.

Part D. How can you see who makes more per hour just by looking at the graphs? Explain.

Part E. Write one equation to represent how much money Kell earns in x hours and one equation to represent how much money Mariko earns in h hours.

Instructional Task 2 (MTR.3.1)

Kelsi works as a lifeguard at the local pool. After an 8-hour day at work, she earns \$100.

Part A. Write an equation that describes the relationship between the number of hours worked and the amount of money that she earns.

Part B. Kelsi would like to earn \$450 to buy a new gaming system. Use your equation to determine how many hours she needs to work to buy a new gaming system.

Instructional Items

Instructional Item 1

Saylor needs to walk her dog Holloway every day after work. After 5 days Saylor walked her dog a total of 200 minutes. Write an equation that describes the relationship between the amount of minutes Saylor walked her dog each day.

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive*